

Homework 12

Please read Section 3.2 of Durrett through the end of Example 3.15. We did not have much time with the class worksheet because of the quiz. The front of the worksheet matches Durrett's Example 3.12, and the back matches his Example 3.14. I recommend completing the worksheet as you read those examples to help you follow his calculations and to practice for the homework questions.

1. Please work Exercise 67 from **Chapter 2**. The answer in the back of the book is $1/2$. Your goal is to explain how to get this value.

A student studies 12 problems from which the professor will randomly choose 6 for a test. If the student can solve 9 of the problems, what is the probability she can solve at least 5 of the problems on the test?

We use the "urn problem" approach. The professor picks 6 problems from the "urn" of 12 problems. The student can solve 9 of those problems. Let X be the random variable representing the number of problems picked that she can solve. We want to calculate $P(X = 5) + P(X = 6)$.

$$P(X = 5) = C_{9,5} C_{3,1} / C_{12,6} = 9 / 22$$

$$P(X = 6) = C_{9,6} / C_{12,6} = 1 / 11$$

$$P(X = 5) + P(X = 6) = 9 / 22 + 1 / 11 = 11 / 22 = 1 / 2.$$

2. Please work Exercise 69 from **Chapter 2**. The answer in the back of the book is 0.5814. Your goal is to explain how to get this value. Reviewing question 3 from the first midterm exam may help you.

We can solve this exercise using the strategy from the first midterm. There are 20^5 ways to pick five children to serve snacks (allowing repeats) for the 5 days of the week (Monday through Friday). There are $P_{20,5}$ ways to pick 5 distinct children for these five days. Assuming all 20^5 possibilities are equally likely, the probability of picking 5 distinct children is $P_{20,5} / 20^5 = 0.5814$ (exactly).

3. Two people are playing bridge as partners. Each gets a hand of 13 cards drawn from a standard deck of 52 cards. The first player's hand has two aces. Given that information, what is the probability that the second player's hand has 0 aces? 1 ace? 2 aces?

This is a conditional probability question. Use the "probability pro" approach: to calculate these probabilities, imagine the second player drawing a hand of 13 cards from the 39 cards remaining after the first player draws.

Probability second player has 0 aces given that first player has 2 aces:

After the first player draws, there are 39 cards including 2 aces. We use the “urn problem” method to calculate the probability. It is $C_{2,0} C_{37,13} / C_{39,13} \approx 0.4385964912$.

Probability second player has 1 ace given that first player has 2 aces:

It is $C_{2,1} C_{37,12} / C_{39,13} \approx 0.4561403509$.

Probability second player has 2 aces given that first player has 2 aces:

It is $C_{2,2} C_{37,11} / C_{39,13} \approx 0.1052631579$.

4. Please work Exercise 15 from **Chapter 3**. Durrett’s answer (from the back of the book) is 0.6. The following steps help you calculate the probability using the two-stage experiment approach.

a. In the first stage 2 letters fall off the sign. Let B_1 be the event “the letters that fall off are different” and B_2 be the event “the letters that fall off are the same.” Calculate the probabilities of these events:

$P(B_1) = 8 / 10$ (any pair except MM or II)

$P(B_2) = 2 / 10$ (MM or II)

b. Let A be the event “after the person puts the letters back, the sign still says ‘MIAMI’.” Calculate the probability of A conditional on each event B_i .

$P(A | B_1) = 1/2$

$P(A | B_2) = 1$

c. Use the law of total probability to calculate $P(A)$.

$P(A) = P(B_1) P(A | B_1) + P(B_2) P(A | B_2) = (8/10)(1/2) + (2/10)(1) = 4/10 + 2/10 = 6/10 = 3/5$.

5. Please work Exercise 16 from **Chapter 3**.

Two balls are drawn from an urn with balls numbered from 1 up to 10. What is the probability that the two numbers will differ by more (>) than 3?

We use the two-stage-experiment method. The first stage is picking the first ball. Let B_i be the event “the first ball is numbered i .” Let A be the event “the two numbers differ by more than 3.”

$$P(A) = P(B_1)P(A | B_1) + P(B_2)P(A | B_2) + \dots + P(B_{10})P(A | B_{10})$$

$P(B_i) = 1/10$ (all i). The conditional probabilities vary depending on i . To calculate these probabilities, for each i , I wrote the numbers 1–10 and counted how many differed from i by more than 3. Given that i is the first draw, each of the nine numbers not equal to i is equally like for the second draw. The following table shows my calculations.

i	1	2	3	4	5	6	7	8	9	10
$P(A B_i)$	6/9	5/9	4/9	3/9	3/9	3/9	3/9	4/9	5/9	6/9

The law of total probability then gives us $P(A) = 42/90 = 7/15$.

6. Please work Exercise 19 from **Chapter 3**. Durrett’s answer from the back of the book is $11/216$. Your goal is to explain how to get that answer.

Durrett’s description of this chance experiment is a bit confusing. You roll two dice and your friend also rolls two dice. If you roll, for example, a (1, 1) (first die 1 and second die 1), then your friend has to roll a (1, 1) to match you. If you roll a (1, 2) (first die 1 and second die 2), though, your friend can match your numbers by rolling a (1, 2) or a (2, 1). The following steps help you calculate the probability that you have the same two numbers using the two-stage experiment approach.

- a. In the first stage, you roll two dice. Let B_1 be the event “you roll two of the same number” and B_2 be the event “you roll two different numbers.” Calculate the probabilities of these events:

$$P(B_1) = 6/36 = 1/6 \text{ (This event is } \{(1,1), (2,2), (3,3), (4,4), (5,5), (6,6)\}.)$$

$$P(B_2) = 5/6 \text{ (} B_2 \text{ is the complement of } B_1.)$$

- b. Let A be the event “you and your friend have the same two numbers.” Calculate the probability of A conditional on each event B_i .

$$P(A | B_1) = 1/36$$

$$P(A \mid B_2) = 2/36$$

c. Use the law of total probability to calculate $P(A)$.

$$P(A) = P(B_1) P(A \mid B_1) + P(B_2) P(A \mid B_2) = (1/6)(1/36) + (5/6)(2/36) = 1/216 + 10/216 = 11/216$$